

AWARE: A Non-Interactive Anonymous Whistleblowing System against Recipient Corruption

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Abstract—Anonymous whistleblowing systems allow users to report misconduct while hiding their identities from recipients and protecting report contents from unauthorized access. Existing systems typically encrypt each report under a recipient public key and carry the submission over Tor. This conceals the user’s network address, but leaves two points of trust. In an organizational setting, the authentication server authenticates users and can correlate token issuance with a later submission. A recipient also holds the decryption key, so its corruption gives an adversary unilateral access to the report. We present *AWARE*, a non-interactive anonymous whistleblowing protocol for registered users (insiders). With *AWARE*, an authenticated user can submit a report anonymously without contacting the authentication server at submission time, and no single recipient can decrypt the report unilaterally. The protocol separates authentication from submission: the authentication server periodically issues pre-tokens to all users in the organization. To submit a report, a user locally finalizes an unused pre-token; the mailbox (the bulletin board) verifies the token’s validity and ownership without learning the user’s identity. A committee of recipients then opens accepted reports jointly through verifiable and accountable threshold opening. We analyze *AWARE*’s correctness and security. Experiments show that *AWARE* has practical cryptographic and communication overheads across reporting, verification, and threshold opening.

Index Terms—whistleblowing system, dependable reporting, unlinkability, holder-bound submission, accountable threshold opening

I. INTRODUCTION

WHISTLEBLOWING is a common way for organizations and government agencies to learn about misconduct. The U.S. Securities and Exchange Commission’s Whistleblower Program received approximately 27,000 tips in fiscal year 2025.¹ The Association of Certified Fraud Examiners reports that tips were the initial detection method in 43% of cases in its 2026 occupational-fraud study.² Dedicated systems [1]–[4] let users, including employees and contractors, report suspected fraud, abuse, safety violations, or other organizational misconduct to designated recipients. Such reports may expose both the whistleblower’s identity and sensitive evidence, so the channel must protect both.

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¹<https://www.sec.gov/files/fy25-annual-whistleblower-report.pdf>

²<https://www.acfe.com/about-the-acfe/newsroom-for-media/press-releases/press-release-detail?s=occupational-fraud-2026-a-report-to-the-nations-pr>

Existing systems typically combine Tor-based submission with encryption under a recipient public key [1]–[3]. A user accesses the submission site through Tor and sends a report, together with any supporting files, to the mailbox. Tor hides the user’s network address from the mailbox. The system stores the report in encrypted form for the designated recipient. The recipient later retrieves the encrypted report and uses the corresponding secret key to decrypt it for review. Encryption protects the stored report contents from parties without the decryption key. Beyond submission and storage, the reporting protocol must address trust in user authentication and report opening. This leads to two challenges.

Challenge 1: report submission without token-issuance correlation. We consider organizational reporting channels that restrict submissions to eligible insiders. Such channels must verify eligibility without disclosing the reporter’s identity. One approach is for an authentication server to check the user’s identity and issue a token for the mailbox to verify. Issuing that token when the user wishes to report creates a record close to the submission time. An authentication server that observes mailbox uploads can compare upload times with its issuance records. Tor or a mix network does not hide the server’s own issuance records. Users therefore need tokens issued independently of reporting and usable at a later time. The submitted token must also remain unlinkable to its issuance record. A copied or transferred token alone must not let another person submit as its owner. The mailbox must thus verify both token validity and ownership without learning the user’s identity. This check must require no contact with the authentication server at submission time.

Challenge 2: report opening without unilateral decryption authority. When one recipient holds the decryption key, its corruption gives an adversary unilateral access to the report. Recipients may themselves be implicated in the misconduct or face conflicting interests. Giving several recipients independent decryption access preserves each recipient’s ability to open the report alone. Even a rule requiring joint approval can be bypassed by a recipient holding the full decryption key. The challenge is to make decryption require contributions from a threshold of distinct recipients while allowing individual participants to behave maliciously. A malicious participant may submit an invalid contribution to disrupt opening. Holding participants accountable requires evidence of who contributed correctly to each accepted report. This evidence must be publicly verifiable without revealing the report contents.

We present *AWARE*, a protocol for anonymous reporting by registered users and accountable threshold opening. The authentication server issues pre-tokens periodically, and users

finalize them locally when submitting encrypted reports. A committee jointly opens accepted reports through publicly verifiable contributions. Our contributions are:

- 1) We combine local token finalization with holder binding to verify ownership while preserving unlinkability to issuance records. We also construct an accountable threshold opening mechanism that binds verifiably encrypted decryption shares to their contributors and the accepted report.
- 2) We integrate these mechanisms into AWARE to support anonymous reporting without contacting the authentication server at submission time. AWARE binds each publicly verified opening contribution to the complete submission accepted by the mailbox. Valid contributions from any t distinct committee members suffice to recover that report.
- 3) We formalize token-issuance correlation and corruption of report recipients in a security model and prove AWARE's security in this model. Experiments with Rust implementation show practical computation and communication costs.

II. RELATED WORK

Prior work identifies anonymity, confidentiality, integrity, plausible deniability, and recipient authenticity as core requirements for whistleblowing systems [30]. Surveys and policy guidance stress the need for protected reporting paths inside organizations [31], [32]. Table I uses these requirements to compare representative technical approaches.

Communication systems. SecureDrop, GlobaLeaks, WikiLeaks, and related deployments use Tor onion services as submission mailboxes [1]–[3], [5]. Onion services hide the service location and the user's network address from the submission service. CenTor [6] improves onion-service performance and security. The proposed SecureDrop Protocol reduces account-creation and message-flow metadata in a proof-of-concept design [4]. Mix-net and private-information-retrieval (PIR) systems [7]–[10] address traffic analysis. Express lowers the cost of metadata-hiding communication [8], XRD scales private messaging [9], and Aimless Onions reduces topology-distribution overhead [10]. These communication mechanisms protect transport metadata. AWARE can use these mechanisms for transport while handling user authentication and report opening at the protocol layer.

Whistleblowing and token systems. Anonymous whistleblowing over authenticated channels is formalized by anonymous transfer [33]. Lower bounds establish limits on non-trivial anonymity in that model [34]. Other reporting schemes use ring signatures, group signatures, proxy signatures, or blockchains [13], [14], [17], [35] to support anonymous reporting or rewards. Ring and group signatures authenticate membership. Blockchain-based schemes support auditing or rewards. These functions address submission and record keeping, not joint control over report decryption. Privacy Pass separates token issuance from token use [11], [12]. Anonymous credentials and keyed-verification credentials [15], [16], [36]–[38] support selective disclosure or anonymous authentication. These primitives provide unlinkable authorization. AWARE integrates token unlinkability with scheduled issuance for

registered users and binds committee opening to accepted reports.

Voting and reputation systems. Electronic-voting protocols [18]–[23] combine privacy with public verifiability through bulletin boards, threshold tallying, and verifiable decryption. Their goals include ballot privacy, receipt-freeness, coercion resistance, tally hiding, and cast-as-intended verification. Anonymous reputation systems [24]–[29] support feedback aggregation, Sybil resistance, and sometimes conditional deanonymization. AWARE applies public verification to the opening of individual confidential reports. Each report is recovered from valid decryption shares provided by t distinct committee members.

III. PRELIMINARIES

A. Notation and Conventions

Let λ denote the security parameter and $\text{negl}(\lambda)$ a negligible function. Algorithms and adversaries are probabilistic polynomial-time (PPT) unless stated otherwise. We write $[n] = \{1, \dots, n\}$ and $\mathbb{G}^* = \mathbb{G} \setminus \{1\}$. The notation $x \xleftarrow{\$} S$ denotes uniform sampling from a finite set S . Exponents are computed modulo the group order. Structured inputs have fixed canonical, type-separated encodings; $\parallel$ denotes concatenation. For the groups below, $H_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1^*$, $H_2 : \{0, 1\}^* \rightarrow \mathbb{G}_2$, and $H_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_p$ denote hash functions. $H_4 : \{0, 1\}^* \rightarrow \{0, 1\}^{2\lambda}$ is collision resistant, and H_5 denotes a key-derivation hash. Hash uses are domain-separated, with tags implicit in the notation.

B. Bilinear Groups

Let $\mathbb{G}_1$ and $\mathbb{G}_2$ be cyclic groups of prime order p , with generators g and $\tilde{g}$, respectively, and let $\mathbb{G}_T$ be a group of the same order. A pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is efficiently computable, non-degenerate, and bilinear: $e(X^a, Y^b) = e(X, Y)^{ab}$ for all $X \in \mathbb{G}_1$, $Y \in \mathbb{G}_2$, and $a, b \in \mathbb{Z}_p$. We use Type-3 groups, with no efficient isomorphism between $\mathbb{G}_1$ and $\mathbb{G}_2$ and no pairing within either source group.

C. Threshold GS Decryption (TGS)

We use Boneh and Shoup's threshold GS encryption construction [39], denoted by TGS, which supports associated data and publicly verifiable decryption shares. Associated data is public and authenticated but not encrypted. The scheme has the following interface:

- $(pk_{th}, pk_c, (sk_i^{th})_{i \in [n]}) \leftarrow \text{DKeyGen}(1^\lambda, n, t)$: generate the threshold public key, public verification material, and one local decryption state for each party. We realize this interface by standard distributed key generation (DKG) using Shamir sharing and verifiable secret sharing [39, Sec. 22.4]. Under the honest, non-aborting setup in Section IV-B, its outputs have the distribution specified by DKeyGen .
- $ct \leftarrow \text{Enc}(pk_{th}, M, ad)$: encrypt M under pk_{th} with associated data ad .
- $b \leftarrow \text{CtVerify}(pk_{th}, ct, ad)$: publicly verify ciphertext well-formedness.

TABLE I
COMPARISON OF REPRESENTATIVE APPROACHES FOR ANONYMOUS REPORTING UNDER THEIR STATED SECURITY MODELS.

Approach	Properties				
	Whistleblower Anonymity ¹	No Unilateral Access ²	Multi-report Unlinkability ³	Report Integrity ⁴	Threshold Opening ⁵
Tor / Onion Services [1], [2], [5], [6]	●	○	○	○	○
Mix / PIR [7]–[10]	●	○	○	○	○
Credential / Token [11]–[17]	●	○	●	●	○
Electronic Voting [18]–[23]	●	●	○	●	●
Anonymous Reputation [24]–[29]	●	○	○	●	○
AWARE (Ours)	●	●	●	●	●

Legend: ● Supported ● Partial / conditional ○ Not supported. **Notes:** ¹ Anonymity follows each scheme's threat model; AWARE assumes periodic issuance and anonymous submission channels. Partial protection covers selected observers or servers. ² No unilateral access requires joint decryption; AWARE requires t valid shares. ³ Unlinkability concerns reports by the same user; AWARE fixes public eligibility conditions and tags. Privacy Pass operates outside registered-insider reporting. ⁴ Integrity concerns well-formed accepted submissions and append-only records, rather than report truthfulness. ⁵ Accountable opening publicly verifies and attributes contributions; any t valid contributions from distinct members recover the same accepted report.

- $d_i \leftarrow \text{ShareDec}(sk_i^{\text{th}}, \text{ct}, \text{ad})$: compute party i 's decryption share, or reject if the ciphertext is invalid.
- $b \leftarrow \text{ShareVrfy}(pk_c, i, \text{ct}, \text{ad}, d_i)$: publicly verify that d_i is a valid share attributed to party i .
- $(M/\perp, \text{blame}) \leftarrow \text{Combine}(pk_c, \text{ct}, \text{ad}, \{d_j\}_{j \in \mathcal{J}})$: combine the input shares and output the plaintext or $\perp$, together with blame information for invalid shares.

We require correctness and threshold AD-only CCA security (Definition 7 in Section VI-D). Correctness requires `Combine` to recover M from every honestly generated ciphertext and any set of at least t honestly generated valid shares. Ciphertext and share verification must also be sound: except with negligible probability, values accepted by `CtVrfy` and `ShareVrfy` satisfy the validity relations of the Boneh–Shoup threshold instantiation.

D. Tagged Non-interactive Blind Signatures (TNIBS)

A tagged non-interactive blind signature (TNIBS) scheme [40] issues a presignature for a recipient public key pk_R , public tag η , and issuance input μ . The recipient uses the corresponding secret key sk_R to finalize the presignature into a signature on a randomized message m , without contacting the signer. The interface is $\text{TNIBS} = (\text{KeyGen}, \text{RKeyGen}, \text{Issue}, \text{Obtain}, \text{Verify})$.

Here μ denotes the issuance input. In AWARE, $\mu = (\nu, \tau)$ contains the issuance nonce ν and epoch position τ , and the finalized message is the pair (ρ, ζ) . Protocol 2 writes ν as $\text{nonce}_{r,\tau}$. The construction uses tag-based equivalence-class signatures (TBEQ) [40].

- $(sk, pk) \leftarrow \text{KeyGen}(1^\lambda)$: generate the signer key pair.
- $(sk_R, pk_R) \leftarrow \text{RKeyGen}(1^\lambda)$: generate the recipient key pair.
- $\text{psig} \leftarrow \text{Issue}(sk, pk_R, \eta, \mu)$: generate a presignature bound to pk_R , public tag η , and issuance input μ .
- $(m, \sigma) \leftarrow \text{Obtain}(sk_R, pk, \text{psig}, pk_R, \eta, \mu)$: finalize the presignature into a signature σ on message m .
- $b \leftarrow \text{Verify}(pk, \eta, (m, \sigma))$: output $b = 1$ if the signature is valid, and $b = 0$ otherwise.

We require (i) one-more unforgeability per public tag, counting randomized signatures on the same (η, m) pair once; (ii) tagged recipient blindness, which hides the recipient public key used to finalize a valid signature under a fixed public tag; and (iii) tagged nonce blindness, which hides the nonce and presignature used to finalize a valid signature under a fixed recipient public key and public tag.

IV. MODELS AND PROBLEM DEFINITION

A. System Model

AWARE involves four types of entities (Fig. 1).

- **Authentication Server ($\mathcal{AS}$)**: The server verifies user eligibility, maintains the mapping between identities and registered public keys, and periodically issues pre-tokens. Issuance is independent of report submission.
- **User ($\mathcal{U}$)**: A registered insider holds a secret key and a pool of pre-tokens. The user locally finalizes a token and submits an encrypted report with a holder binding proof. A submitting user is denoted by $\mathcal{W}$ (the whistleblower).
- **Bulletin Board ($\mathcal{BB}$)**: The submission mailbox (bulletin board) validates submissions and records the first valid use of each token serial number in a public append-only ledger. It also records opening contributions with authenticated committee-member attribution.
- **Committee Members ($\{\mathcal{CM}_i\}_{i=1}^n$)**: Designated report recipients hold threshold-decryption states and contribute verifiably encrypted shares for accepted reports. Valid contributions from any t distinct members for the same accepted record suffice to recover the report.

B. Threat Model

We analyze protocol transcripts with report submissions sent over anonymous channels, abstracting away network addresses, paths, and communication timing.

Authentication server. The server is semi-honest and follows registration and issuance over authenticated channels. It knows users' identities, registered keys, pre-tokens, and issuance

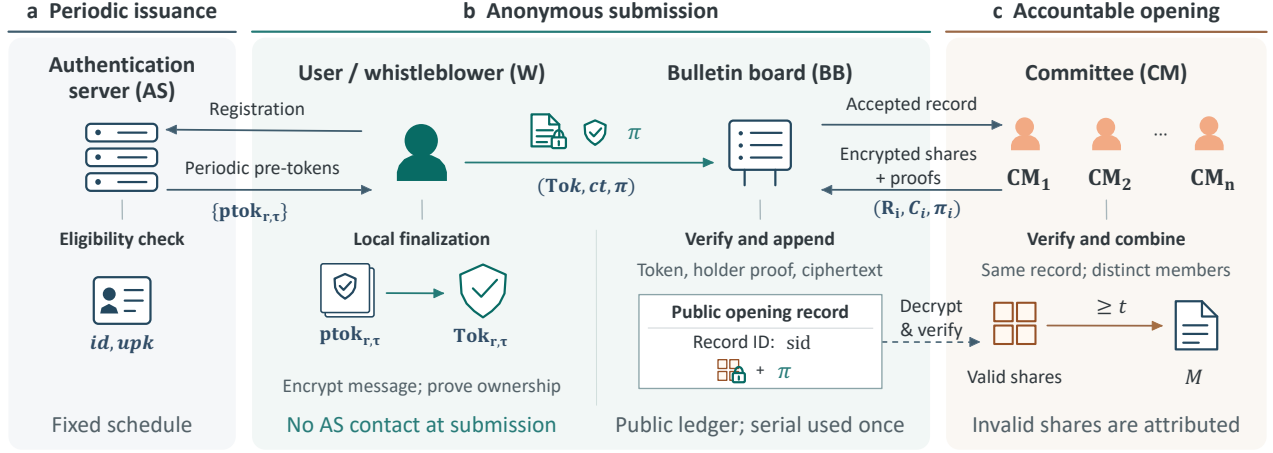


Fig. 1. System model of AWARE. (a) The authentication server checks user eligibility and periodically issues pre-tokens. (b) The user finalizes a token locally and submits an encrypted report with a holder binding proof; the bulletin board verifies and records the submission. (c) The committee recovers the report from verified opening contributions of t distinct members bound to the same accepted-record identifier sid . Here, r denotes the issuance epoch and $\tau \in [m]$ the user's pre-token slot within that epoch.

Protocol 1: System Initialization and Registration.

(1) **Setup**($1^\lambda, n, t$) Executed once during initialization.

Step 1. Require $2 \leq t \leq n$. Run TNIBS.KeyGen for message length 3: sample $(a_1, a_2, a_3) \leftarrow (\mathbb{Z}_p^*)^3$, set $\text{ask} = (a_1, a_2, a_3)$, and publish $\text{apk} = (A_1, A_2, A_3) = (\tilde{g}^{a_1}, \tilde{g}^{a_2}, \tilde{g}^{a_3})$.

Step 2. Run $(pk_{\text{th}}, pk_c, (sk_i^{\text{th}})_{i \in [n]}) \leftarrow \text{TGS.DKeyGen}(1^\lambda, n, t)$ using DKG. Let $f \in \mathbb{Z}_p[X]$ be the resulting Shamir polynomial of degree at most $t-1$, with $\alpha = f(0)$ and $\alpha_i = f(i)$. Set $\Theta = g^\alpha$, $\Theta_i = g^{\alpha_i}$, and $\tilde{\Theta}_i = \tilde{g}^{\alpha_i}$. Give $sk_i^{\text{th}} = (i, \alpha_i, \Theta_i, \tilde{\Theta}_i)$ only to CM_i ; publish pk_{th} , containing Θ , and pk_c , containing $(\Theta_i, \tilde{\Theta}_i)_{i \in [n]}$.

Step 3. Establish a shared decryption key $z \leftarrow \mathbb{Z}_p^*$, publish the share-encryption public key $P = g^z$, and initialize the $\mathcal{BB}$ ledger. Every committee member receives z .

(2) **RegU**(1^λ) Executed locally by an eligible user.

Step 1. Run TNIBS.RKeyGen : sample $x \leftarrow \mathbb{Z}_p^*$, set $\text{usk} = x$, $\text{upk} = g^x$, and register upk after eligibility verification.

Protocol 2: Periodic Token Issuance and Finalization.

(1) **RegIss**($\text{ask}, \{\text{upk}\}, \eta_r, m$) Executed by $\mathcal{AS}$ for issuance epoch r .

Step 1. Use a fresh issuance tag η_r . For each registered upk and each $\tau \in [m]$, sample $\text{nonce}_{r,\tau}, y_{r,\tau} \leftarrow \mathbb{Z}_p^*$, and compute

$$h_{r,\tau} = H_1(\text{nonce}_{r,\tau}), \quad \psi_{r,\tau} = H_1(\eta_r, \tau), \quad T_{r,\tau} = H_2(\eta_r)^{y_{r,\tau}^{-1}}, \\ Z_{r,\tau} = (\text{upk}^{a_1} h_{r,\tau}^{a_2} \psi_{r,\tau}^{a_3})^{y_{r,\tau}}, \quad Y_{r,\tau} = g^{y_{r,\tau}^{-1}}, \quad \tilde{Y}_{r,\tau} = \tilde{g}^{y_{r,\tau}^{-1}}.$$

Set $\text{psig}_{r,\tau} = (Z_{r,\tau}, Y_{r,\tau}, \tilde{Y}_{r,\tau}, T_{r,\tau})$.

Step 2. Deliver $\text{ptok}_{r,\tau} = (\eta_r, \text{nonce}_{r,\tau}, h_{r,\tau}, \text{psig}_{r,\tau})$ for $\tau \in [m]$, with padding independent of reporting.

(2) **RegObt**($\text{usk}, \text{ptok}_{r,\tau}$) Executed locally by the user.

Step 1. Parse $\text{psig}_{r,\tau} = (Z_{r,\tau}, Y_{r,\tau}, \tilde{Y}_{r,\tau}, T_{r,\tau})$, set $x = \text{usk}$ and $\psi_{r,\tau} = H_1(\eta_r, \tau)$, and sample $\omega_{r,\tau} \leftarrow \mathbb{Z}_p^*$. Compute

$$\rho_{r,\tau} = h_{r,\tau}^{x^{-1}}, \quad \zeta_{r,\tau} = \psi_{r,\tau}^{x^{-1}}, \quad Z'_{r,\tau} = Z_{r,\tau}^{x^{-1} \omega_{r,\tau}}, \\ Y'_{r,\tau} = Y_{r,\tau}^{\omega_{r,\tau}^{-1}}, \quad \tilde{Y}'_{r,\tau} = \tilde{Y}_{r,\tau}^{\omega_{r,\tau}^{-1}}, \quad T'_{r,\tau} = T_{r,\tau}^{\omega_{r,\tau}^{-1}}.$$

Step 2. Store $\text{Tok}_{r,\tau} = (\eta_r, \rho_{r,\tau}, \zeta_{r,\tau}, \sigma_{r,\tau})$ together with its local index τ , where $\sigma_{r,\tau} = (Z'_{r,\tau}, Y'_{r,\tau}, \tilde{Y}'_{r,\tau}, T'_{r,\tau})$.

randomness. It attempts to link public submissions to these records, possibly with corrupted committee members.

Users. User keys are generated honestly during registration. Registered users may then be malicious. They know their own keys and tokens and attempt to exceed the issued token count. An outsider may obtain an honest user's token and attempt to

submit a report as its owner or replay an earlier submission. Users protected by anonymity and unlinkability remain honest.

Bulletin board. The board is trusted to enforce acceptance, append-only order, and committee attribution. It publishes accepted records and opening contributions, which adversaries can inspect and replay in attempts to gain repeated acceptance.

Protocol 3: Report Generation and Authentication.

(1) RepGen($pk_{th}, M, Tok_{r,\tau}$) *Executed locally by the user.*

Step 1. Choose an unused finalized $Tok_{r,\tau} = (\eta_r, \rho_{r,\tau}, \zeta_{r,\tau}, \sigma_{r,\tau})$ and set $ad = Tok_{r,\tau}$.

Step 2. Run $TGS.Enc(pk_{th}, M, ad)$ using Θ from pk_{th} : sample $\beta \leftarrow \mathbb{Z}_p^*$ and compute the ciphertext $ct = (v, \tilde{v}, c_M, \bar{w})$ as

$$\begin{aligned} v &= g^\beta, & \tilde{v} &= \tilde{g}^\beta, & w &= \Theta^\beta, & k &= H_5(v, w), \\ c_M &\leftarrow E_s(k, M), & \bar{u} &= H_1(v, ad), & \bar{w} &= \bar{u}^\beta. \end{aligned}$$

Step 3. Set $x = usk$ and $\psi_j = H_1(\eta_r, j)$. For $j \neq \tau$, sample $c_j, z_j \leftarrow \mathbb{Z}_p$ and set $R_j = \zeta_{r,\tau}^{z_j} \psi_j^{-c_j}$; sample $r_\pi \leftarrow \mathbb{Z}_p$ and set $R_\tau = \zeta_{r,\tau}^{r_\pi}$. Compute

$$c = H_3(Tok_{r,\tau}, ct, R_1, \dots, R_m), \quad c_\tau = c - \sum_{j \neq \tau} c_j, \quad z_\tau = r_\pi + c_\tau x \pmod{p}.$$

Set $\pi = ((c_j, z_j))_{j \in [m]}$ and submit $(Tok_{r,\tau}, ct, \pi)$ to $\mathcal{BB}$ for PostAccept.

(2) wAuth($apk, Tok_{r,\tau}$) *Executed by $\mathcal{BB}$.*

Step 1. Parse $Tok_{r,\tau} = (\eta_r, \rho_{r,\tau}, \zeta_{r,\tau}, \sigma_{r,\tau})$; reject if η_r is outside the public acceptance window or $\rho_{r,\tau}, \zeta_{r,\tau} \notin \mathbb{G}_1^*$.

Step 2. Parse $\sigma_{r,\tau} = (Z'_{r,\tau}, Y'_{r,\tau}, \tilde{Y}'_{r,\tau}, T'_{r,\tau})$. Output 1 iff all three equations hold:

$$e(Z'_{r,\tau}, \tilde{Y}'_{r,\tau}) = e(g, A_1)e(\rho_{r,\tau}, A_2)e(\zeta_{r,\tau}, A_3), \quad e(Y'_{r,\tau}, \tilde{g}) = e(g, \tilde{Y}'_{r,\tau}), \quad e(g, T'_{r,\tau}) = e(Y'_{r,\tau}, H_2(\eta_r)).$$

(3) PostAccept($Tok_{r,\tau}, ct, \pi; \mathcal{S}$) *Executed by $\mathcal{BB}$.*

Step 1. Parse $\pi = ((c_j, z_j))_{j \in [m]}$. For each $j \in [m]$, compute $\psi_j = H_1(\eta_r, j)$ and $R_j = \zeta_{r,\tau}^{z_j} \psi_j^{-c_j}$. Check that $\rho_{r,\tau} \notin \mathcal{S}$ for the spent-serial set $\mathcal{S}$, and verify

$$wAuth(apk, Tok_{r,\tau}) = 1, \quad TGS.CtVerfy(pk_{th}, ct, Tok_{r,\tau}) = 1, \quad \sum_{j=1}^m c_j = H_3(Tok_{r,\tau}, ct, R_1, \dots, R_m) \pmod{p}.$$

Step 2. If all checks pass, compute $sid = H_4(Tok_{r,\tau}, ct, \pi)$, atomically append the accepted record $rec = (sid, Tok_{r,\tau}, ct, \pi)$ to the ledger, and set $\mathcal{S} \leftarrow \mathcal{S} \cup \{\rho_{r,\tau}\}$; otherwise leave both unchanged.

Committee. Committee setup is honest: DKG uses authenticated private channels and authenticated broadcast. The adversary chooses a static corruption set before setup and learns those members' states afterward. Corrupted members may deviate arbitrarily from the protocol. They know their threshold-decryption states and the shared decryption key, which lets them recover decryption shares from published opening contributions. They attempt early report decryption or disrupt opening through invalid or withheld opening contributions.

C. Security Goals

AWARE provides the following security properties.

Whistleblower anonymity. An accepted token hides its link to the registered user and issuance record. Multi-report unlinkability also hides whether reports originate from the same user at the protocol layer.

Token-based authentication. Holder binding prevents misuse of an exposed token under the conditions above. Token unforgeability bounds accepted submissions by issued pre-tokens, and each token serial number is accepted at most once.

Report confidentiality. Report contents remain confidential while the adversary holds valid shares from fewer members than the opening threshold, given public metadata and length.

Accountable threshold opening. Each public contribution can be verified and attributed to its sender. Verified contributions bind valid encrypted shares to one accepted record. Any t such contributions from distinct members suffice to recover the report. Invalid contributions can be detected and attributed to their senders.

V. AWARE PROTOCOL

A. Overview of AWARE

AWARE proceeds through the following four phases.

Initialization. Protocol 1 generates the server's TNIBS signing key. The committee runs DKG to obtain threshold secret shares, an encryption public key, and share-verification keys. It separately establishes z and $P = g^z$ to encrypt opening shares for publication.

Registration and token issuance. To separate eligibility checks from reporting, RegIss periodically issues m pre-tokens per registered user. We extend Hanzlik's TNIBS [40] with an epoch-position coordinate so that each pre-token occupies a distinct issuance slot. The user finalizes it locally through RegObt, using the secret key and signature randomization to hide its issuance record (Protocol 2). The issuance group and token acceptance window determine the candidate users for a submission; smaller groups and narrower windows reduce the anonymity set.

Reporting and authentication. To protect report contents, RepGen in Protocol 3 encrypts the report under the committee's threshold encryption public key. It uses the selected token as associated data to bind authentication to the ciphertext. To make redemption require the registered holder's secret, we construct a Schnorr OR proof over the issuance slots. It binds the token to the report ciphertext while hiding the selected position. The bulletin board verifies token validity, holder binding, and ciphertext well-formedness. To prevent repeated acceptance, PostAccept atomically records the first valid submission for each token serial number. It assigns an accepted-

Protocol 4: Accountable Threshold Opening.

(1) RepDec(pk_{th}, sk_i^{th}, rec) Executed by committee member $\mathcal{CM}_i$.

Step 1. Parse $rec = (sid, Tok_{r,\tau}, ct, \pi)$, check that the record appears in the accepted ledger and that $sid = H_4(Tok_{r,\tau}, ct, \pi)$. Set $ad = Tok_{r,\tau}$ and run $TGS.CtVrfy(pk_{th}, ct, ad)$. Parse $ct = (v, \tilde{v}, c_M, \bar{w})$, set $\bar{u} = H_1(v, ad)$, and require $e(v, \tilde{g}) = e(g, \tilde{v})$ and $e(\bar{w}, \tilde{g}) = e(\bar{u}, \tilde{v})$.

Step 2. Run $TGS.ShareDec(sk_i^{th}, ct, ad)$. Using α_i from sk_i^{th} , compute $w_i = v^{\alpha_i}$ and $d_i = (i, w_i)$. The corresponding verification key is $\tilde{\Theta}_i$ in pk_c .

Step 3. Sample $r_i, k_i \leftarrow \mathbb{Z}_p^*$, and compute

$$R_i = g^{r_i}, \quad C_i = w_i P^{r_i}, \quad A = e(P, \tilde{g}), \quad U_i = g^{k_i}, \quad V_i = A^{k_i}.$$

Set $ctx_i = (i, sid, ct, Tok_{r,\tau})$, and compute

$$c_i = H_3(\text{open}, ctx_i, R_i, C_i, U_i, V_i), \quad s_i = k_i + c_i r_i \pmod{p}, \quad \pi_i = (c_i, s_i).$$

Here open is the opening-proof domain label; all fields use canonical encoding. The accepted ledger fixes the committee keys and P for this context. Submit $(sid, i, R_i, C_i, \pi_i)$ with authenticated attribution to $\mathcal{CM}_i$. Using the public checks in RepCom Step 2, $\mathcal{BB}$ atomically admits the first opening-proof-valid contribution for each (sid, i) to the active set $\mathcal{O}_{sid}$; rejected attempts may be retained separately for audit.

(2) RepCom($pk_{th}, pk_c, z, P, rec, \mathcal{O}_{sid}$) Executed by the committee during opening.

Step 1. Parse $rec = (sid, Tok_{r,\tau}, ct, \pi)$ and check its ledger membership and record identifier. Set $ad = Tok_{r,\tau}$ and run $TGS.CtVrfy(pk_{th}, ct, ad)$. Parse $ct = (v, \tilde{v}, c_M, \bar{w})$, compute $\bar{u} = H_1(v, ad)$, and require $e(v, \tilde{g}) = e(g, \tilde{v})$ and $e(\bar{w}, \tilde{g}) = e(\bar{u}, \tilde{v})$. Initialize $\mathcal{J} \leftarrow \emptyset$ and blame $\leftarrow \emptyset$.

Step 2. Process entries $(sid_j, j, R_j, C_j, \pi_j)$ from $\mathcal{O}_{sid}$ in ledger order. Ignore entries with $j \notin [n]$, $sid_j \neq sid$, inconsistent authenticated attribution, or $j \in \mathcal{J}$. Parse $\pi_j = (c_j, s_j)$, set $ctx_j = (j, sid, ct, Tok_{r,\tau})$, and compute

$$A = e(P, \tilde{g}), \quad B_j = e(C_j, \tilde{g})e(v, \tilde{\Theta}_j)^{-1}, \quad U'_j = g^{s_j} R_j^{-c_j}, \quad V'_j = A^{s_j} B_j^{-c_j}.$$

Accept the proof iff $c_j = H_3(\text{open}, ctx_j, R_j, C_j, U'_j, V'_j)$. If verification fails, add $\mathcal{CM}_j$ to blame and ignore the entry.

Otherwise recover $w_j = C_j R_j^{-z}$, set $d_j = (j, w_j)$, and run $TGS.ShareVrfy(pk_c, j, ct, ad, d_j)$, which accepts iff

$e(w_j, \tilde{g}) = e(v, \tilde{\Theta}_j)$. If share verification succeeds, add j to $\mathcal{J}$; otherwise, add $\mathcal{CM}_j$ to blame.

Step 3. If $|\mathcal{J}| < t$, publish blame and return $(\perp, \text{blame})$. Otherwise, recover the TGS decryption value from any t -subset $\mathcal{J}_t \subseteq \mathcal{J}$ as follows:

$$\lambda_j = \prod_{\ell \in \mathcal{J}_t \setminus \{j\}} \frac{\ell}{\ell - j}, \quad w = \prod_{j \in \mathcal{J}_t} w_j^{\lambda_j}.$$

Derive $k = H_5(v, w)$. Publish blame and return (M, blame) , where $M = D_s(k, c_M)$.

record identifier sid that binds the token, ciphertext, and holder binding proof for subsequent opening.

Accountable threshold opening. To make contributions publicly checkable while hiding shares, we augment Boneh-Shoup's robust TGS [39] with share encryption and a Chaum-Pedersen-style consistency proof bound to the complete accepted record. In Protocol 4, RepDec checks the record and ciphertext, then encrypts and proves its share. The authenticated ledger attributes each contribution. For each record, the bulletin board admits each member's first proof-valid contribution to an active set of size at most n . RepCom verifies the proofs, recovers shares using z , and applies ShareVrfy; any t valid shares suffice to decrypt.

B. Construction of AWARE

Use separate H_1 domains for nonces, epoch positions, and ciphertexts, and separate H_3 domains for holder binding and opening proofs. Model the first two H_1 domains as independent random oracles into $\mathbb{G}_1^*$. Let E_s, D_s denote the symmetric encryption and decryption algorithms in the Boneh-Shoup instantiation, and use H_5 for key derivation. Check group

membership on parsing and enforce the specified non-identity conditions on ρ, ζ , keys, and TGS elements.

VI. SECURITY ANALYSIS

A. Correctness

Theorem 1 (Correctness). *Except with negligible probability of nonce or serial-number collisions, every honestly generated fresh submission is accepted by PostAccept, and any t honest opening contributions for the same accepted record recover the original report.*

Proof sketch. For $\rho = h^{x^{-1}}$ and $\zeta = \psi_{r,\tau}^{x^{-1}}$, direct substitution gives the three equations checked by WAAuth, and the honestly generated TGS ciphertext passes CtVrfy. Since $\psi_{r,\tau} = \zeta^x$,

$$\zeta^{z\tau} \psi_{r,\tau}^{-c\tau} = \zeta^{r\pi + c\tau x} (\zeta^x)^{-c\tau} = \zeta^{r\pi},$$

so the holder proof is accepted. For honest contributions, $w_i = v^{\alpha_i}$, and any t distinct shares satisfy $\prod_{i \in \mathcal{J}_t} w_i^{\lambda_i} = v^\alpha = \Theta^\beta$, yielding the same H_5 key as RepGen and the original report M . $\square$

B. Whistleblower Anonymity

Anonymity hides the association between an accepted submission and its registered user or issuance record in the protocol view.

Protocol views. Let $\text{View}_{\text{AS}}^{\text{priv}}$ contain the issuer keys (ask, apk), tag policy, identity–public-key registry, and ordered issuance records ($\text{upk}, \eta_r, \tau, \text{nonce}_{r,\tau}, y_{r,\tau}, \text{ptok}_{r,\tau}$). Let View_{BB} contain accepted records ($\text{sid}, \text{Tok}, \text{ct}, \pi$), attributed contributions ($\text{sid}, i, R_i, C_i, \pi_i$), and ledger order. For a static set $\mathcal{C} \subseteq [n]$, $|\mathcal{C}| < t$, write

$$\text{View}_{\mathcal{C}} = (\kappa_{\mathcal{C}}, \{sk_i^{\text{th}}\}_{i \in \mathcal{C}}), \quad \kappa_{\mathcal{C}} = \begin{cases} z, & \mathcal{C} \neq \emptyset, \\ \perp, & \mathcal{C} = \emptyset. \end{cases}$$

The combined view is $\text{View}^{\text{anon}} = (\text{View}_{\text{AS}}^{\text{priv}}, \text{View}_{\text{BB}}, \text{View}_{\mathcal{C}})$.

Users in the same eligibility class receive m pre-tokens per epoch, independently of report submission. The games fix registration, issuance, and non-challenge protocol interactions across the two histories.

Definition 1 (Single-report anonymity). Fix $\mathcal{C} \subseteq [n]$, $|\mathcal{C}| < t$, before setup. Given the combined view above, including ask , a PPT adversary chooses two histories that differ only in the honest user supplying the challenge token. They have the same payload, public epoch/tag, ledger position, eligibility conditions, issuance sequence, and non-challenge interactions. Both candidate records are unfinalized and reserved for the challenge, and their tokens are valid and unused. The challenger samples a uniform bit b , realizes history $\mathcal{H}_b$ using Protocols 1–3, and returns its updated combined view. Experiment $\text{Anon}_{\mathcal{A}}$ returns 1 iff the adversary guesses b . Anonymity requires

$$\text{Adv}_{\mathcal{A}}^{\text{Anon}}(\lambda) := |\Pr[\text{Anon}_{\mathcal{A}}(\lambda) = 1] - 1/2|$$

to be negligible for every such adversary.

Definition 2 (Multi-report unlinkability). AWARE satisfies multi-report unlinkability if, for every polynomial $q = q(\lambda)$ and every PPT adversary $\mathcal{A}$, the following advantage is negligible:

$$\text{Adv}_{\mathcal{A}}^{\text{MAnon}}(\lambda) := |\Pr[\text{MAnon}_{\mathcal{A}}(\lambda) = 1] - \frac{1}{2}|$$

Extend Definition 1 to q challenge slots with the same ordered payloads, public epoch/tags, and ledger positions. Candidate users for each slot have the same public eligibility conditions. The histories differ only in the assignment of slots to issuance records, allowing both same-user and different-user assignments. Each token is valid and unused, and each issuance record occurs at most once per history. The challenger realizes $\mathcal{H}_b$ for uniform b , returns the resulting view of the q submissions, and accepts iff the adversary guesses b .

Definition 3 (Multi-instance issuer-side TNIBS blindness). TNIBS satisfies multi-instance issuer-side blindness if every PPT adversary has negligible advantage in this multi-record version of Hanzlik’s recipient- and nonce-blindness games.

The challenger reveals (ask, apk) while keeping user secret keys hidden. Before and after the challenge, $\mathcal{B}$ may request honestly generated user public keys, issuance for registered users, and auxiliary finalizations. Issuance queries specify a user, tag, and position. Repeated user–tag–position queries

are rejected; valid queries return complete ordered issuance records, including pre-tokens and issuer randomness. Finalization rejects unknown, previously finalized, or challenge-locked records. Auxiliary and challenge records may share user keys. The nonce and epoch-position domains of H_1 are independent random oracles, with direct and issuance queries sharing the same nonce oracle.

At challenge time, $\mathcal{B}$ outputs equal-length vectors $\mathbf{I}_0 = (I_{0,1}, \dots, I_{0,q})$ and $\mathbf{I}_1 = (I_{1,1}, \dots, I_{1,q})$ from the issuance ledger. Each record must be unfinalized and occur at most once per vector; both vectors must have identical ordered tags. Otherwise the challenger aborts. Corresponding positions τ may differ; records may overlap across vectors, and user keys may repeat. The challenger locks all records in both vectors, samples $b \xleftarrow{\$} \{0, 1\}$, finalizes the records of $\mathbf{I}_b$, and returns the ordered token vector. After further oracle queries, $\mathcal{B}$ outputs $\hat{b}$. The experiment returns 1 iff $\hat{b} = b$. Its view includes (ask, apk), the ordered pre- and post-challenge oracle transcript, and the challenge vector. Define

$$\text{Adv}_{\mathcal{B}, \text{TNIBS}}^{\text{MSIBnd}}(\lambda) := |\Pr[\text{MSIBnd}_{\mathcal{B}, \text{TNIBS}}(\lambda) = 1] - \frac{1}{2}|.$$

Theorem 2 (Multi-instance blindness of the extended Hanzlik TNIBS construction). Consider the natural length-three TBEQ extension of Hanzlik’s TNIBS construction on (upk, h, ψ) in Protocols 1–4, in the random oracle model. Write iDDH for distinguishing $(g^x, g^{x^{-1}})$ from (g^x, g^z) , and write sDDH for distinguishing $(g^\alpha, g^\beta, g^{\beta^{-1}}, g^{\alpha\beta^{-1}})$ from $(g^\alpha, g^\beta, g^{\beta^{-1}}, g^\gamma)$, where all exponents are uniform in $\mathbb{Z}_p^*$. Let u bound the number of accepted recipient-key queries, and let $v \leq 2q$ bound the number of distinct records in $\mathbf{I}_0 \cup \mathbf{I}_1$ whose nonce relation is replaced below. Let $Q_{H_1}^{\text{N}}$ bound the total direct queries and issuance calls to the shared type-separated TNIBS-nonce table. Let $Q_{H_1}^{\text{P}}$ bound the direct and protocol-induced distinct queries to the epoch-position table of H_1 . Then there are PPT reductions $\mathcal{D}_1, \dots, \mathcal{D}_u$ and $\mathcal{E}_1, \dots, \mathcal{E}_v$ and $\mathcal{F}_1, \dots, \mathcal{F}_v$ such that

$$\begin{aligned} \text{Adv}_{\mathcal{B}}^{\text{MSIBnd}}(\lambda) &\leq \sum_{i=1}^u \text{Adv}_{\mathcal{D}_i}^{\text{iDDH}}(\lambda) + Q_{H_1}^{\text{N}} \sum_{j=1}^v \text{Adv}_{\mathcal{E}_j}^{\text{sDDH}}(\lambda) \\ &\quad + u Q_{H_1}^{\text{P}} \sum_{j=1}^v \text{Adv}_{\mathcal{F}_j}^{\text{sDDH}}(\lambda) + \epsilon_{\text{nonce}}(\lambda), \end{aligned}$$

where $\epsilon_{\text{nonce}} \leq (Q_{H_1}^{\text{N}})^2 / |\mathcal{N}_\lambda|$ accounts for a sampled nonce repeating a prior TNIBS-nonce input, with $\mathcal{N}_\lambda = \mathbb{Z}_p^*$. Hence the construction satisfies Definition 3 under these assumptions.

Theorem 3 (Single-report anonymity). Assume the iDDH and sDDH assumptions of Theorem 2 hold in $\mathbb{G}_1$ and H_3 is modeled as a random oracle. Assume anonymous submission channels and the common issuance conditions in Section VI-B. Then AWARE satisfies Definition 1 for the combined static view.

Theorem 4 (Multi-report unlinkability). Under the assumptions of Theorem 3, AWARE satisfies multi-report unlinkability against the static combined view in Definition 2.

Proof sketch. We first establish TNIBS blindness, then reduce both privacy properties to that result. Hybrid indices are local to each part.

TNIBS blindness. We remove the recipient, nonce, and issuance-position relations in turn.

Hybrid G_0 . Run the real blindness experiment with a shared TNIBS-nonce table and an independent epoch-position table for H_1 . For a finalized record, write $H_1(N) = g^{r_N}$, $H_1(\eta, \tau) = g^{q_{\eta, \tau}}$, and $W_i = g^{x_i^{-1}}$. Its coordinates are $\rho = W_i^{r_N}$ and $\zeta = W_i^{q_{\eta, \tau}}$.

Hybrid G_1 . Abort if a sampled nonce repeats a prior nonce-table input. The winning-probability gap is bounded by $\varepsilon_{\text{nonce}}$.

Hybrids $G_{2,i}$, $i = 0, \dots, u$. Starting from $G_{2,0} = G_1$, replace one recipient's inverse-key base W_i by an independent uniform non-identity element. Use that base for all of its challenge and auxiliary finalizations. Given either base, sample $s \leftarrow \mathbb{Z}_p^*$ and generate the signature as

$$Y' = g^s, \quad \tilde{Y}' = \tilde{g}^s, \quad T' = H_2(\eta)^s, \quad Z' = (g^{a_1} \rho^{a_2} \zeta^{a_3})^{s^{-1}}.$$

For fixed issuer randomness y , the bijection $s = (y\omega)^{-1}$ gives the real joint distribution of the finalized signature and issuance record. Each adjacent transition is bounded by the corresponding iDDH advantage, with issuance records fixed.

Hybrids $G_{3,j}$, $j = 0, \dots, v$. Starting from $G_{3,0} = G_{2,u}$, replace one distinct challenge record's ρ by an independent uniform non-identity element and regenerate its signature by the same formula. An sDDH embedding guesses the record's nonce-table entry. For each batch k , sample independent $c_k \leftarrow \mathbb{Z}_p^*$. For the target record, set $W_k = (g^{\beta^{-1}})^{c_k}$, program $H_1(N^*) = g^\alpha$, and set $\rho^* = (g^\gamma)^{c_k}$; this is the preceding distribution when $\gamma = \alpha\beta^{-1}$ and is independent uniform otherwise. The guessing loss is $Q_{H_1}^N$ for each transition.

Hybrids $G_{4,j}$, $j = 0, \dots, v$. Starting from $G_{4,0} = G_{3,v}$, replace one challenge record's ζ in the same way. For the guessed recipient i^* , sample $c \leftarrow \mathbb{Z}_p^*$ and set $W_{i^*} = (g^{\beta^{-1}})^c$, program $H_1(\eta^*, \tau^*) = g^\alpha$, and set $\zeta^* = (g^\gamma)^c$; the two sDDH cases give the preceding and independent distributions. The embedding guesses both the recipient-key query and the epoch-position entry, giving loss $uQ_{H_1}^P$ per transition.

In $G_{4,v}$, challenge coordinates are independent of their recipients and issuance positions. Admissibility fixes the public tags and permitted auxiliary views, so the two final token-vector distributions are identical. Summing the adjacent distinguishing bounds and the nonce-abort probability under the theorem's advantage normalization gives the stated bound.

Anonymity and unlinkability. For one challenge report, set $q = 1$; for unlinkability, use the ordered vector of q reports.

Hybrid G_0 . Run the history-zero experiment.

Hybrid G_1 . Simulate each holder binding proof by sampling $c_j, z_j \leftarrow \mathbb{Z}_p$, setting $R_j = \zeta^{z_j} H_1(\eta, j)^{-c_j}$, and programming the holder-binding H_3 domain to return $\sum_j c_j$. Since $\zeta \neq 1$, the commitments are unpredictable. The special-HVZK simulator and the Fiat-Shamir theorem [39, Thm. 20.3] bound the error by $Q_P(Q_P + Q_3)/p$ for Q_P proofs and Q_3 oracle queries.

Hybrid G_2 . Switch the token vector using TNIBS blindness. The two histories reserve unfinalized records with the same ordered tags, so they give admissible challenge vectors. Encrypt each fixed payload with the returned token as associated data and simulate its proof as above. This is randomized polynomial-time post-processing; the ciphertext domain of H_1

is independent of the domains used in the blindness reduction. The reduction generates the threshold state locally, including exposed member states, and answers issuance and permitted finalization queries through the TNIBS interface.

Hybrid G_3 . Restore the real holder binding proofs. The same simulation bound yields history one. Admissibility fixes issuance, payloads, public tags, ledger positions, and non-challenge interactions, so

$$\begin{aligned} \text{Adv}_A^{\text{Anon}}(\lambda) &\leq \text{Adv}_{B_1}^{\text{MSIBnd}}(\lambda) + \text{negl}(\lambda), \\ \text{Adv}_A^{\text{MAnon}}(\lambda) &\leq \text{Adv}_{B_q}^{\text{MSIBnd}}(\lambda) + \text{negl}(\lambda). \end{aligned}$$

Both bounds are negligible by Theorem 2, establishing single-report anonymity and multi-report unlinkability. $\square$

C. Token-Based Authentication

Holder binding protects token ownership; token unforgeability bounds accepted submissions by issued pre-tokens. Acceptance follows `PostAccept` in Protocol 3, including its atomic update of the spent-serial set $\mathcal{S}$.

Definition 4 (Holder binding). *The adversary receives an honest user's valid unused token $\text{Tok}^* = (\eta^*, \rho^*, \zeta^*, \sigma^*)$ and the public and issuer-side information allowed by the model. Other users' secret keys may be exposed, but x^* and valid holder binding proofs for Tok^* remain hidden. AWARE satisfies holder binding if the probability of producing (ct^*, π^*) that passes the holder binding check in `PostAccept` is negligible.*

Theorem 5 (Holder binding). *The construction satisfies holder binding in the random oracle model under the discrete-logarithm assumption in $\mathbb{G}_1$.*

Definition 5 (Token unforgeability). *The issuer's signing key and issuance state remain secret. A PPT adversary receives the public parameters and epoch schedule, and may register users through `RegU`, obtaining their honestly generated key pairs. It may request `RegIss` for a registered user, active tag, and slot in $[m]$, at most once per user-tag-slot tuple. Let q_{iss} count answered issuance queries; local finalization leaves this count unchanged. Starting with an empty spent set, process the adversary's ordered submission list through `PostAccept` and let N_{acc} count accepted submissions. Experiment TokUnf_A returns 1 iff $N_{\text{acc}} > q_{\text{iss}}$. Token unforgeability requires*

$$\text{Adv}_A^{\text{TokUnf}}(\lambda) := \Pr[N_{\text{acc}} > q_{\text{iss}}]$$

to be negligible for every such adversary.

Definition 6 (TNIBS one-more unforgeability). *We use Han-zlik's TNIBS one-more game [40] restricted to honestly generated recipient keys, matching AWARE's registration model. The adversary receives apk , obtains registered recipient key pairs from $\mathcal{O}_{\text{RKeyGen}}$, and requests presignatures through $\mathcal{O}_{\text{TNIBS.Issue}}(i, \eta, \mu)$ for registered users. Let q_η count answered queries under tag η . It wins by producing, for some η^* , more than q_{η^*} valid signatures on distinct pairs (ρ, ζ) . Randomizations of a signature on the same pair count once. One-more unforgeability requires its winning probability $\text{Adv}_{B, \text{TNIBS}}^{\text{OM-Unf}}(\lambda)$ to be negligible for every PPT adversary.*

Concrete one-more unforgeability. Assume EUF-CMA security of the length-three TBEQ instantiation over $(\mathbb{G}_1^*)^3$. Each accepted token gives a valid TBEQ signature on (g, ρ, ζ) . The canonical-representative reduction of Hanzlik [40, Thm. 4] extends directly to length three: fixing the first coordinate to g makes distinct (ρ, ζ) pairs distinct TBEQ equivalence classes. Thus TBEQ EUF-CMA security implies Definition 6 in the random oracle model.

Theorem 6 (Token unforgeability). *If the underlying TNIBS scheme satisfies one-more unforgeability as in Definition 6, then AWARE satisfies token unforgeability with honestly generated user keys and an uncompromised issuer signing key.*

Proof sketch. Holder binding follows by extracting a witness from an accepting proof; token unforgeability follows by reducing excess accepted tokens to a TNIBS forgery.

Holder binding. *Hybrid G_1 :* embed the discrete logarithm. Given $X = g^x$, set $\text{upk}^* = X$, sample nonzero δ_j , and program $H_1(\eta^*, j) = X^{\delta_j}$ for every branch. Set $\zeta^* = g^{\delta_{\tau^*}}$; program $H_1(N^*) = X^b$ and set $\rho^* = g^b$ for random nonzero b . Generate the finalized signature with the distribution used in the blindness proof. The reduction can simulate the target token while retaining the unknown x in every branch relation.

Hybrid G_2 : fork the accepting query. Rewind at the accepting query using the general forking lemma [41, Lemma 1]. For success probability ϵ and $q_3 \geq 1$ queries, let $\epsilon_q \geq \max\{\epsilon - 1/p, 0\}$ exclude success at an unqueried point. Two accepting transcripts with identical commitments and distinct challenges occur with probability at least $\max\{0, \epsilon_q(\epsilon_q/q_3 - 1/p)\}$.

Conclusion of the fork. Some branch has $c_j \neq c'_j$. Its two verification equations yield

$$x' = (z_j - z'_j)(c_j - c'_j)^{-1}, \quad H_1(\eta^*, j) = (\zeta^*)^{x'}.$$

Together with $H_1(\eta^*, j) = X^{\delta_j}$ and $\zeta^* = g^{\delta_{\tau^*}}$, this gives $x = x' \delta_{\tau^*} \delta_j^{-1} \pmod{p}$. Thus any non-negligible holder-binding success yields a non-negligible algorithm for discrete logarithm, contradicting the DL assumption.

Token unforgeability. Use the TNIBS challenger's public key as apk, obtain registered keys from its recipient-key oracle, and answer every valid issuance query under tag η while maintaining its count q_η . Every token accepted by PostAccept contains a valid TNIBS signature, while the atomic spent-serial rule makes the accepted ρ values, and hence the pairs (ρ, ζ) , distinct. If

$$N_{\text{acc}} = \sum_{\eta} |J_\eta| > q_{\text{iss}} = \sum_{\eta} q_\eta,$$

then some tag η^* satisfies $|J_{\eta^*}| > q_{\eta^*}$. The accepted signed pairs under η^* therefore yield a one-more TNIBS forgery, so

$$\text{Adv}_{\mathcal{A}}^{\text{TokUnf}}(\lambda) \leq \text{Adv}_{\mathcal{B}, \text{TNIBS}}^{\text{OM-Unf}}(\lambda).$$

The right-hand side is negligible by TNIBS one-more unforgeability, establishing token unforgeability. $\square$

D. Report Confidentiality

Report confidentiality protects the plaintext when the combined issuance view and corrupted states reveal challenge shares

from fewer than t members. The reduction uses threshold AD-only CCA security with $\text{ad} = \text{Tok}$.

Definition 7 (Threshold AD-only CCA security). *Use the threshold game of Boneh and Shoup [39, Game 22.5]. Before $D\text{KeyGen}$, the adversary chooses (n, t) and a static set $\mathcal{L}$ of $t-1$ members; it receives the public and verification keys and their secret shares. In world b , each encryption query (M_0, M_1, ad) with equal-length messages returns $\text{Enc}(pk_{\text{th}}, M_b, \text{ad})$. A share query returns all n decryption shares only for associated data absent from earlier encryption queries. Security requires*

$$\text{Adv}_{\mathcal{A}, \text{TGS}}^{\text{ThADCCA}}(\lambda) := |\Pr[\hat{b} = 1 \mid b = 0] - \Pr[\hat{b} = 1 \mid b = 1]|$$

to be negligible for every PPT adversary.

Boneh and Shoup [39, Thm. 22.7] derive this threshold GS security from AD-only strong-CCA security of GS encryption.

Definition 8 (Report confidentiality). *AWARE satisfies report confidentiality if, for every PPT adversary $\mathcal{A}$, the following advantage is negligible:*

$$\text{Adv}_{\mathcal{A}}^{\text{Conf}}(\lambda) := |\Pr[\text{Conf}_{\mathcal{A}}(\lambda) = 1] - \frac{1}{2}|$$

The adversary first fixes $\mathcal{L} \subseteq \mathcal{T} \subseteq [n]$, $|\mathcal{T}| < t$, exposing member states in $\mathcal{L}$ and allowing challenge shares indexed by $\mathcal{T}$. It receives public parameters and $\text{View}_{\mathcal{L}}^{\text{conf}} = (\text{View}_{\text{AS}}^{\text{priv}}, \text{View}_{\text{BB}}, \text{View}_{\mathcal{L}})$, as in Section VI-B. Before and after the challenge, it may obtain honestly issued tokens, submit through PostAccept, and request opening contributions for accepted records, subject to Definition 7. Records and contributions enter the ledger in order; repeated member-record queries return the same share. Any member corruption reveals z , hence the underlying published shares.

The adversary chooses equal-length (M_0, M_1) and an unused honestly finalized Tok^ from the challenger's token outputs, with no prior answered share query under that token. For uniform b , the challenger runs RepGen on M_b using the retained holder secret and issuance position, processes the result through PostAccept, and returns the updated view. Both worlds fix the token, ledger position, public metadata, issuance view, and corrupted states.*

For the exact challenge record, the experiment directly reveals the underlying valid decryption share d_i only for $i \in \mathcal{T} \setminus \mathcal{L}$, which strengthens the adversary relative to receiving the encrypted Protocol 4 contribution. Shares indexed by $\mathcal{L}$ are already available from the exposed states. No share is returned for $i \notin \mathcal{T}$ or for a modified ciphertext under Tok^ . Repeated indices count once, so at most $|\mathcal{T}| < t$ distinct challenge shares are available. The experiment returns 1 iff the adversary's guess $\hat{b}$ equals b .*

Theorem 7 (Report confidentiality). *Assume the DKG realizes TGS.DKeyGen under the honest-setup model, and the Boneh-Shoup component in RepGen satisfies Definition 7 with $\text{ad} = \text{Tok}$, and H_3 is modeled as a random oracle. Then AWARE satisfies Definition 8 against the static combined view $\text{View}_{\mathcal{L}}^{\text{conf}}$ and every static $\mathcal{L} \subseteq \mathcal{T}$ with $|\mathcal{T}| < t$. More precisely, for a suitable PPT reduction $\mathcal{R}$,*

$$\text{Adv}_{\mathcal{A}}^{\text{Conf}}(\lambda) \leq \frac{1}{2} \text{Adv}_{\mathcal{R}, \text{TGS}}^{\text{ThADCCA}}(\lambda) + \text{negl}(\lambda).$$

Proof sketch. We compare the plaintext worlds in Definition 8 through the following hybrids.

Hybrid G_0 . Run the real experiment with challenge plaintext M_0 , including the DKG state and allowed opening queries.

Hybrid G_1 . Use the threshold state supplied by the Boneh–Shoup challenger. Choose $\mathcal{L}' \supseteq \mathcal{T}$ with $|\mathcal{L}'| = t - 1$, expose only states in $\mathcal{L}$, and retain the others. The honest DKG gives the TGS.DKeyGen distribution. Generate TNIBS keys, user secrets, issuance records, and z locally, retaining the challenge token’s holder secret and issuance position.

Hybrid G_2 . Replace the encryption of M_0 by the challenger’s encryption of M_1 , keeping Tok^* as associated data. Generate the real holder proof using the retained witness and the returned ciphertext; this requires no knowledge of the challenge bit. Recompute the record identifier. For permitted non-challenge records, obtain shares from the AD-only CCA oracle and encrypt them with locally sampled opening randomness. For challenge queries in $\mathcal{T} \setminus \mathcal{L}$, compute shares from the retained states. All exposed challenge shares have indices in $\mathcal{T}$, with $|\mathcal{T}| < t$; the simulation forwards only associated-data values allowed by the underlying game. This transition is bounded by the threshold AD-only CCA advantage; Boneh and Shoup [39, Thm. 22.7] supply the underlying threshold GS reduction.

Hybrid G_3 . Restore the honest DKG distribution to obtain the real experiment for M_1 . The endpoint probabilities and the two-world normalization yield

$$\text{Adv}_{\mathcal{A}}^{\text{Conf}}(\lambda) \leq \frac{1}{2} \text{Adv}_{\mathcal{R}, \text{TGS}}^{\text{ThADCCA}}(\lambda) + \text{negl}(\lambda).$$

Threshold AD-only CCA security makes this advantage negligible and establishes report confidentiality. $\square$

E. Accountable Threshold Opening

Definition 9 (Accountable threshold opening). *AWARE provides accountable threshold opening if the authenticated ledger attributes each accepted opening contribution to a committee member and public verification binds its valid encrypted decryption share to an accepted-record identifier sid , except with negligible probability. Any t such contributions from distinct members with the same sid suffice for report recovery by TGS.Combine for a correctly generated report ciphertext. For an arbitrary accepted ciphertext, they yield its prescribed TGS decryption result, which may be $\perp$. Public share hiding additionally requires that honestly generated encrypted contributions and their proofs be simulatable from public keys and accepted records alone, for observers without z or committee secret states.*

Theorem 8 (Accountable threshold opening). *Assume collision resistance of H_4 , the random oracle model for H_3 , correctness of the stated TGS initialization and symmetric encryption, and the prime-order pairing setting used by the construction. If $\mathcal{BB}$ provides authenticated attribution and accepted-record membership, AWARE satisfies Definition 9, with public share hiding under DDH in $\mathbb{G}_1$ in the public type-3 pairing setting.*

Proof sketch. We exclude record collisions and false proofs, then reconstruct from the remaining valid shares.

Hybrid G_0 . Run public contribution verification and RepCom for an accepted record rec with identifier sid , using the bulletin

board’s active set, which contains at most one opening-proof-valid contribution per member.

Hybrid G_1 . Abort if two distinct accepted records have the same identifier. Such an event produces an H_4 collision. Otherwise, every entry checked against sid refers to the fixed record, and authenticated attribution identifies the contributor.

Hybrid G_2 . Additionally abort if public verification accepts a false opening-proof statement. Use A and B_j from Protocol 4 and fix the commitments (U, V) . Two accepting responses with distinct challenges satisfy

$$g^s R_j^{-c} = g^{s'} R_j^{-c'}, \quad A^s B_j^{-c} = A^{s'} B_j^{-c'}.$$

Setting $r_j = (s - s')(c - c')^{-1}$ yields $R_j = g^{r_j}$ and $B_j = A^{r_j}$. This establishes special soundness for the common-exponent relation in $\mathbb{G}_1$ and $\mathbb{G}_T$, both of order p . The standard Fiat–Shamir soundness bound, including repeated verification attempts [39, Sec. 20.3.6], gives $\Pr[\text{Bad}_{\text{proof}}] \leq (Q_3 + Q_V)/p$, where Q_3 counts opening-domain queries and Q_V counts verification attempts at both the bulletin board and RepCom. Each remaining accepted entry has a witness r_j satisfying

$$R_j = g^{r_j}, \quad e(C_j, \tilde{g}) = e(v, \tilde{\Theta}_j) e(P, \tilde{g})^{r_j}.$$

With $P = g^z$, recovering $w_j = C_j R_j^{-z}$ gives

$$e(w_j, \tilde{g}) = e(C_j, \tilde{g}) e(R_j, \tilde{g})^{-z} = e(v, \tilde{\Theta}_j).$$

Non-degeneracy in the prime-order groups identifies this as the unique share $w_j = v^{\alpha_j}$, which passes ShareVrfy.

Conclusion in G_2 . For any t distinct verified indices, Shamir interpolation gives $\sum_j \lambda_j \alpha_j = \alpha$, hence $\prod_j w_j^{\lambda_j} = v^\alpha$. This is the TGS decryption value. For a correctly generated report ciphertext, symmetric decryption recovers the report; for an arbitrary accepted ciphertext, Combine returns its prescribed decryption result, including $\perp$. Let Bad_{open} denote acceptance of an incorrectly bound or invalid opening contribution. For a collision-finding adversary $\mathcal{B}$, the preceding hybrids give

$$\Pr[\text{Bad}_{\text{open}}] \leq \text{Adv}_{H_4, \mathcal{B}}^{\text{CR}}(\lambda) + \frac{Q_3 + Q_V}{p}.$$

For polynomial Q_3, Q_V , this bound is negligible under the stated assumptions, establishing contribution validity and threshold recovery.

For public share hiding, let q count honest contributions and compare public views through hybrids S_0, S_1, S_2 . In S_0 , contributions are real. In S_1 , simulate each opening proof by sampling $c, s \leftarrow \mathbb{Z}_p$, setting $U = g^s R_i^{-c}$, $V = A^s B_i^{-c}$, and programming the opening-domain H_3 query to c . Fiat–Shamir zero knowledge bounds this change by $O(q(q + Q_3)/p)$ [39, Thm. 20.3]. In S_2 , replace each mask P^{r_i} by an independent uniform $Z_i \in \mathbb{G}_1$, retaining the simulated proofs. Each replacement is a DDH hybrid: embed (P, R_i, D) , set $C_i = w_i D$, and generate threshold states locally to compute w_i and the public verification keys. The final $C_i = w_i Z_i$ is uniform, so the view can be simulated using public records and keys alone. For DDH distinguishing gaps ϵ_i ,

$$\Delta_{\text{pub}} \leq \sum_{i=1}^q \epsilon_i + O(q(q + Q_3)/p).$$

This bound is negligible for polynomially many contributions and oracle queries, establishing public share hiding. $\square$

VII. PERFORMANCE EVALUATION

A. Experimental Setup

We implement AWARE in Rust with MIRACL Core BN462³ for Type-3 pairings and OpenSSL for AES-256-GCM, with security parameter $\lambda = 128$. All AWARE benchmarks use release builds. Communication costs are measured from canonically serialized messages. Unless otherwise specified, we use length-three TNIBS, $m = 15$ tokens per epoch, a committee of $n = 5$ members with threshold $t = 3$, and a 1 MiB payload. The source code is available at <https://github.com/Zora-G/AWARE>. Client operations run on a MacBook Air equipped with an Apple M2 processor and 24 GB of unified memory, running macOS 15.6.1. AS, BB, and CM operations run on an Ubuntu 24.04.4 server equipped with two Intel Xeon Gold 6133 processors (40 physical cores and 80 logical CPUs in total) and 1 TiB of RAM.

B. Computation and Communication Costs

The measured computation costs remain acceptable for the reporting workflow: at the default parameters, RepGen takes 14.78 ms and CompleteOpen takes 120.91 ms. Figure 2 shows where AWARE spends computation. A larger token budget moves more work into the scheduled preparation phase. During report generation, the holder binding proof adds group operations whose cost is independent of payload size for fixed m . Larger payloads amortize this cost, while encryption and challenge hashing over the complete ciphertext require work proportional to payload size. Opening cost is governed mainly by the threshold t , because each valid contribution must be verified and combined; increasing n with the same t has little effect.

The added communication is small relative to report payloads and remains acceptable for deployments that already support document upload. The report payload accounts for almost all submission traffic. At the default 1 MiB payload, the serialized AWARE metadata adds only 0.27% (Table II). The main advance communication cost comes from scheduled issuance: for the measured epoch-tag length, one epoch sends $B_{\text{epoch}} = 621mN_U$ bytes from the authentication server to users. Thus, larger m buys more locally usable tokens at the price of higher scheduled issuance traffic. Opening traffic grows with t , because each selected committee member sends one fixed-size contribution.

C. Scalability and Ablation

AWARE supports batch token issuance and concurrent report acceptance at the measured scale. Figure 3 shows two server-side scaling trends. Parallel issuance reduces epoch time sharply at first, with limited additional gain beyond four AS workers. BB throughput rises with client concurrency until the acceptance path reaches its measured capacity; at

TABLE II
MEAN OPERATION LATENCY, COMMUNICATION, ROLE, AND INVOCATION FREQUENCY AT THE DEFAULT CONFIGURATION.

Operation	Role	Comp. (ms)	Comm. (KiB)	Frequency
Setup	Server	57.06	—	once
RegU	Client	0.40	0.09	once/user
RegIss	AS	94.30	9.10	per epoch
RegObt	Client	50.77	—	per epoch
RepGen	Client	14.78	1,026.75	per report
PostAccept	BB	66.80	—	per report
RepDec	CM	16.92	0.37	per member
RepCom	CM	84.63	—	per opening
CompleteOpen	CMs+CM	120.91	1.10	per opening

Message flows. RegU: $U \rightarrow AS$; RegIss: $AS \rightarrow U$; RepGen: $W \rightarrow BB$; RepDec/CompleteOpen: $CM(s) \rightarrow BB$. CompleteOpen measures total elapsed time for t parallel RepDec calls followed by RepCom, including task coordination. Setup measures local key-generation computation with committee initialization simulated on one host, excluding network communication.

TABLE III
MEAN LATENCY AND COMMUNICATION ADDED BY EACH SECURITY EXTENSION; UNITS ARE SHOWN PER ROW. Δ = AWARE – BASELINE.

Metric	Baseline	AWARE	Δ
<i>Holder binding</i>			
RepGen (client, ms)	2.77	14.78	12.01
PostAccept (BB, ms)	39.02	67.75	28.73
Submission $W \rightarrow BB$ (MiB)	1.00	1.00	< 0.01
<i>Accountable threshold opening</i>			
RepDec (CM, ms)	12.78	16.72	3.94
RepCom (CM, ms)	32.24	82.73	50.49
CompleteOpen (ms)	71.92	119.93	48.01
Contribution $CM \rightarrow BB$ (KiB)	0.07	0.37	0.30

Baseline: the same length-three implementation with holder binding proofs disabled for the first block and plain threshold opening for the second.

higher concurrency, it fluctuates around the plateau rather than increasing monotonically. The reported throughput includes token verification and record acceptance.

The added security mechanisms have clear cost centers. Table III compares AWARE with matched configurations that disable one mechanism at a time. Holder binding mainly affects the client and BB acceptance path, because each submission carries a proof of token ownership. Accountable opening mainly affects RepCom, where encrypted contributions are verified and combined. The added work publicly ties each valid contribution to its committee member and accepted record.

D. Comparison

This comparison isolates the cost of adding anonymous insider authentication and accountable committee opening to a deployed whistleblowing workflow. Table IV compares the cryptographic report-preparation and opening costs of AWARE and GlobalLeaks on the same host. AWARE performs registered-user authentication, holder binding, replay prevention, and accountable threshold opening, whereas GlobalLeaks provides independent recipient access. This extra verification makes AWARE slower for preparation and for small-report opening. As payloads grow, fixed protocol costs are amortized, and

³<https://github.com/miracl/core>

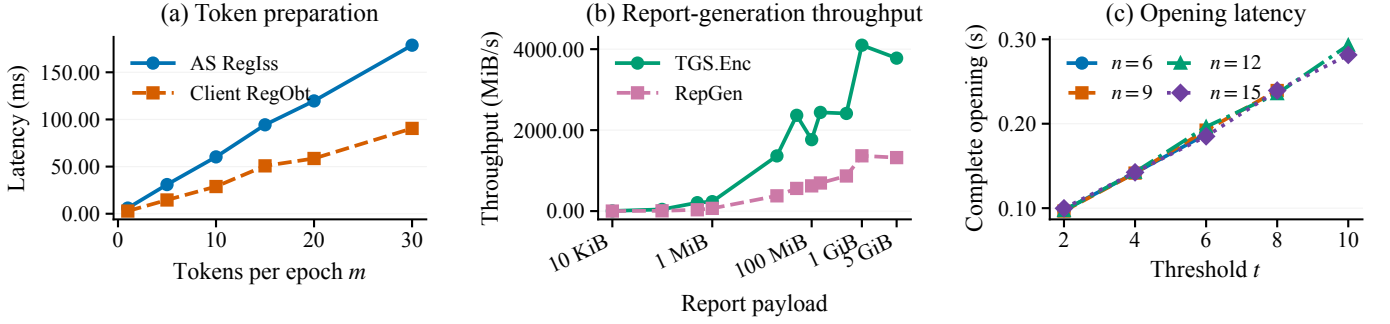


Fig. 2. Protocol scaling. (a) Issuance (Xeon) and finalization (M2) latency versus tokens per epoch. (b) Report-generation and TGS encryption throughput versus payload size on M2. (c) Complete-opening latency versus threshold on Xeon, with one curve per committee size n . Points show mean latency in (a) and (c), and payload size divided by mean latency in (b). Lower latency and higher throughput indicate better performance.

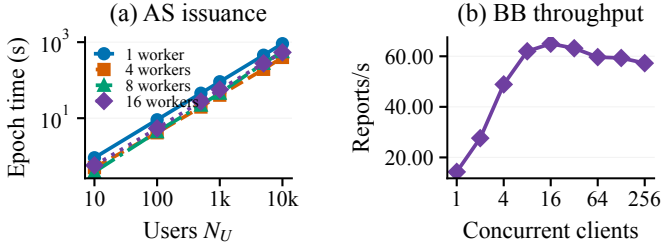


Fig. 3. Server scalability on Xeon. (a) Mean epoch-issuance time versus user population for different AS worker counts; lower is better. (b) Mean accepted reports per second versus concurrent clients with 40 BB workers; higher is better. Lines connect measured means.

AWARE’s combined opening cost is lower from 100 MiB onward. The comparison uses five GlobaLeaks recipients, with opening measured for one recipient. In Table IV, AWARE Opening is the sum of the independently measured mean computation times of one RepDec call and one RepCom call. All entries report measured means.

TABLE IV

MEAN REPORT-PREPARATION AND OPENING COMPUTATION TIME (s) BY PAYLOAD SIZE FOR AWARE AND GLOBALEAKS 5.0.99 ON THE SAME XEON HOST. LOWER IS BETTER; < 0.01 DENOTES LATENCY BELOW 0.01 s.

Payload	Report preparation (s)		Opening (s)	
	AWARE	GlobaLeaks	AWARE	GlobaLeaks
10 KiB	0.03	< 0.01	0.10	< 0.01
1 MiB	0.04	< 0.01	0.10	< 0.01
100 MiB	0.74	0.27	0.14	0.20
1 GiB	8.45	2.82	0.57	2.10
5 GiB	41.19	13.96	3.25	9.86

VIII. CONCLUSION

AWARE enables anonymous reporting by eligible insiders while placing report decryption under accountable committee control. Periodic pre-token issuance separates authentication from the decision to report; local finalization and holder binding provide anonymous redemption tied to the user’s secret key. Threshold encryption and verifiable encrypted contributions jointly bind report recovery to valid shares from distinct members and the accepted record. The security analysis establishes anonymity, unlinkability, unforgeability, confidentiality, and accountable opening under the stated model.

The Rust evaluation shows practical computation and communication costs and quantifies the additional work required for holder binding and accountable opening. Together, these results provide a basis for organizational reporting channels that preserve reporter privacy, distribute decryption authority, and make committee contributions publicly auditable.

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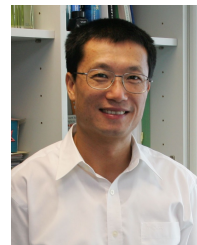
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